

3.3 Quadratic Functions

Many relationships encountered in mathematics, science, engineering, economics, and data analysis involve a variable raised to the power of two. Such relationships are described by **quadratic equations**.

Definition

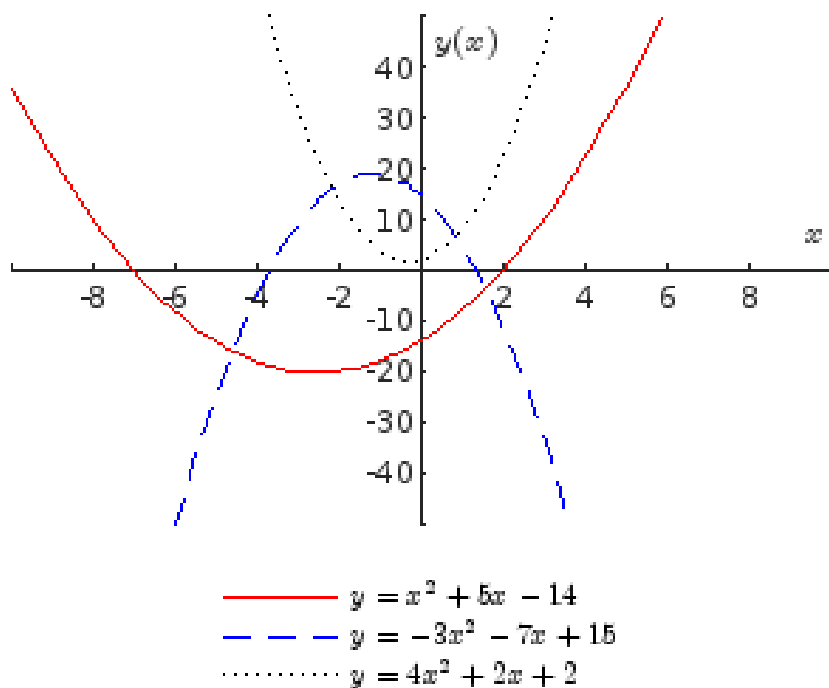
The **quadratic equation** is an equation that can be written in the form

$$y = ax^2 + bx + c,$$

where a , b , and c are constants. $a \neq 0$ and x is the independent variable.

3.3.1 The Graph of a Quadratic Function

The graph of a quadratic equation is called a **parabola**.



Definition

A parabola is a smooth curve whose shape and orientation depend on the coefficients of a , b and c .

- What is the orientation?
 - If $a > 0$, the parabola opens upwards.
 - If $a < 0$, the parabola opens downwards.
- Is it broad or narrow compared to $y = x^2$?
 - $|a| > 1$: narrower.
 - $|a| < 1$: broader.
- The turning point of the curve is called the **vertex**. Where this is depends on the coefficients:
 - Positive c will push it up.
 - Negative c will push it down.
 - Positive or negative b will push it down if $a > 0$ (up if $a < 0$).
 - $b = 0$: on the y -axis.
 - $b > 0$: left of the y -axis if $a > 0$ (right if $a < 0$).
 - $b < 0$: right of the y -axis if $a > 0$ (left if $a < 0$).

The constant c is the y -intercept, as in the linear case (if $x = 0$ then the equation becomes $y = a \times 0^2 + b \times 0 + c = c$).

The curve can cross the x -axis (at $y = 0$) twice, once (just touching it) or never. If it does cross the x -axis, we can calculate the values of x where this occurs by solving:

$$ax^2 + bx + c = 0$$

The **solutions** of $0 = ax^2 + bx + c$ are the same as the x -intercepts of $y = ax^2 + bx + c$ and are also known as the **roots** of $ax^2 + bx + c$.

Definition

The roots of a quadratic equation are the x -coordinates of the points where the corresponding parabola intersects the x -axis.

3.3.2 Solving by Factorisation

Some quadratic equations can be solved by expressing the quadratic expression as a product of two factors.

Worked Example

Solve

$$x^2 + 5x + 6 = 0$$

We seek two numbers whose product is 6 and whose sum is 5.

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

Therefore,

$$(x + 2)(x + 3) = 0$$

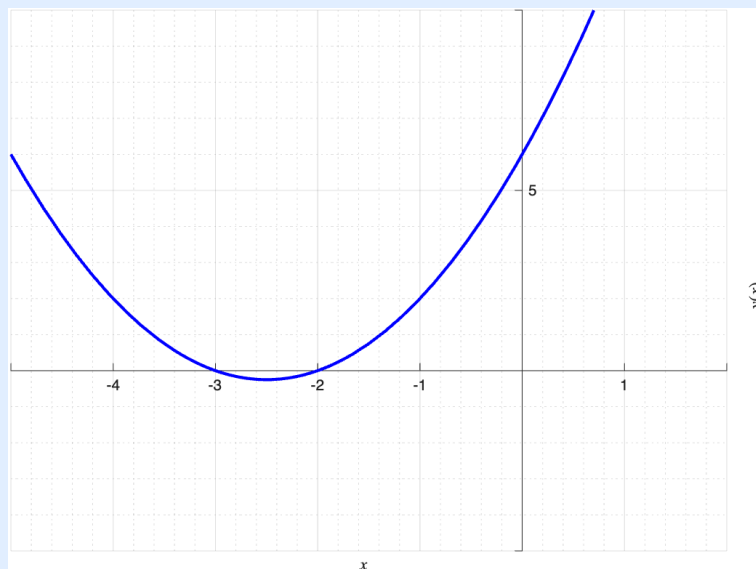
A product is zero if at least one factor is zero. Hence

$$x + 2 = 0 \quad \text{or} \quad x + 3 = 0$$

Therefore,

$$x = -2 \quad \text{or} \quad x = -3$$

The graph looks as follows:



Worked Example

Solve

$$x^2 - x - 20 = 0$$

We seek two numbers whose product is -20 and whose sum is -1 .

$$x^2 - x - 20 = (x - 5)(x + 4)$$

Therefore,

$$(x - 5)(x + 4) = 0$$

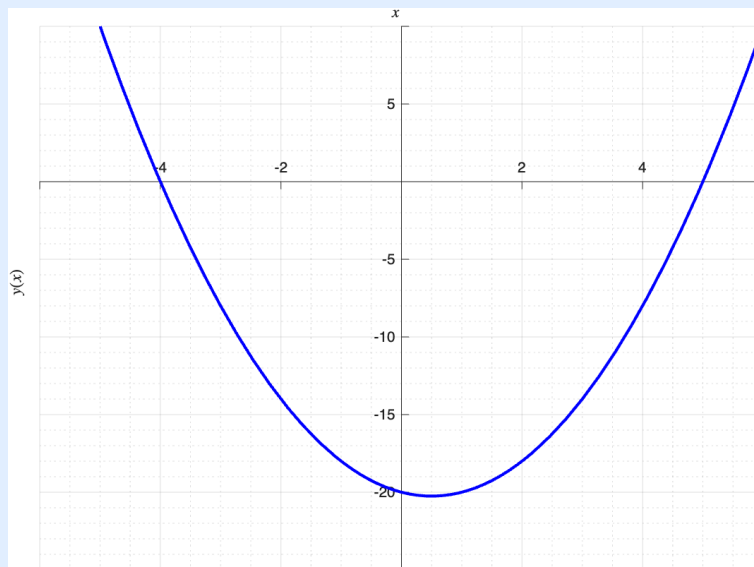
A product is zero if at least one factor is zero. Hence

$$x - 5 = 0 \quad \text{or} \quad x + 4 = 0$$

Therefore,

$$x = 5 \quad \text{or} \quad x = -4$$

The graph looks as follows:



3.3.3 The Quadratic Formula

Not all quadratic equations factorise easily. The quadratic formula can be used to solve any quadratic equation.

Definition

Given

$$ax^2 + bx + c = 0$$

the solutions are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Worked Example

Solve

$$2x^2 + 3x - 2 = 0$$

Here, $a = 2$, $b = 3$ and $c = -2$. Substituting into the quadratic formula gives

$$\begin{aligned} x &= \frac{-3 \pm \sqrt{3^2 - 4(2)(-2)}}{2(2)} \\ &= \frac{-3 \pm \sqrt{25}}{4} \\ &= \frac{-3 \pm 5}{4} \end{aligned}$$

Hence,

$$x = \frac{1}{2} \quad \text{or} \quad x = -2$$

Worked Example

Solve

$$x^2 - 11x + 28 = 0$$

Here, $a = 1$, $b = -11$ and $c = 28$. Substituting into the quadratic formula gives

$$\begin{aligned}x &= \frac{-(-11) \pm \sqrt{(-11)^2 - 4(1)(28)}}{2(1)} \\&= \frac{11 \pm \sqrt{9}}{2} \\&= \frac{11 \pm 3}{2}\end{aligned}$$

Hence,

$$x = 4 \quad \text{or} \quad x = 7$$

3.3.3.1 Factorisation from the Solutions

If there are two distinct, real roots x_1 and x_2 to the quadratic $y = ax^2 + bx + c$, then it is possible to re-write the quadratic in the factorised form:

$$y = a(x - x_1)(x - x_2)$$

However, if there are real and repeated roots, $x_1 = x_2$, then it is possible to re-write the quadratic in the form:

$$y = a(x - x_1)^2$$

3.3.4 The Discriminant

The expression under the square root sign in the quadratic formula has special significance.

Definition

For the quadratic equation $ax^2 + bx + c = 0$, the quantity

$$\Delta = b^2 - 4ac$$

is called the **discriminant**. The discriminant determines the number and type of solutions.

- If $\Delta > 0$, the equation has two distinct real roots.
- If $\Delta = 0$, the equation has one repeated real root.
- If $\Delta < 0$, the equation has no real roots.

Definition

The number of real roots corresponds to the number of intersections between the parabola and the x -axis.

Graph Behaviour	Nature of Roots
Crosses the x -axis twice	Two distinct real roots
Touches the x -axis once	One repeated real root
Does not meet the x -axis	No real roots, i.e. complex roots

3.3.5 Applications of Quadratic Equations

Quadratic models occur naturally in many practical situations.

Engineering Application

Projectile Motion

Ignoring air resistance, the height of a projectile can often be modelled by

$$h(t) = at^2 + bt + c,$$

where t represents time. To determine when the projectile reaches the ground, we solve

$$h(t) = 0$$

This requires finding the roots of a quadratic equation.

Engineering Application

Revenue and Profit Models

Quadratic functions frequently arise in economic and business models. For example,

$$P(x) = -2x^2 + 80x - 300$$

may represent profit as a function of production level x . Analysing the graph allows us to identify the production level that maximises profit.

3.3.6 Exercises

Exercise

1. Solve the following equations by factorisation where possible.

(a) $x^2 + 7x + 12 = 0$

(b) $x^2 - 9x + 20 = 0$

(c) $x^2 + 8x + 7 = 0$

(d) $x^2 + 4x - 5 = 0$

2. Use the quadratic formula to solve:

(a) $2x^2 + 7x + 3 = 0$

(b) $3x^2 - 2x - 8 = 0$

3. Determine the nature of the roots using the discriminant.

(a) $x^2 + 2x + 5 = 0$

(b) $x^2 - 6x + 9 = 0$

(c) $2x^2 + x - 4 = 0$

4. A ball is projected vertically and its height is modelled by

$$h(t) = -5t^2 + 20t + 25$$

(a) Determine the time when the ball reaches the ground.

(b) Sketch the graph.

3.3 Solutions to Exercise 3.3.6

1. Solve the following quadratic equations by factorisation.

- (a) $x^2 + 7x + 12 = 0$. We require two numbers whose product is 12 and whose sum is 7. These numbers are 3 and 4.

$$x^2 + 7x + 12 = 0$$

$$(x + 3)(x + 4) = 0$$

Therefore,

$$x + 3 = 0$$

or

$$x + 4 = 0$$

$$x = -3$$

$$x = -4$$

Therefore, the solutions are $x = -3$ and $x = -4$.

- (b) $x^2 - 9x + 20 = 0$. We require two numbers whose product is 20 and whose sum is -9 . These numbers are -4 and -5 .

$$x^2 - 9x + 20 = 0$$

$$(x - 4)(x - 5) = 0$$

Therefore,

$$x - 4 = 0$$

or

$$x - 5 = 0$$

$$x = 4$$

$$x = 5$$

Therefore, the solutions are $x = 4$ and $x = 5$.

- (c) $x^2 + 8x + 7 = 0$. We require two numbers whose product is 7 and whose sum is 8. These numbers are 1 and 7.

$$x^2 + 8x + 7 = 0$$

$$(x + 1)(x + 7) = 0$$

Therefore,

$$x + 1 = 0$$

or

$$x + 7 = 0$$

$$x = -1$$

$$x = -7$$

Therefore, the solutions are $x = -1$ and $x = -7$.

- (d) $x^2 + 4x - 5 = 0$. We require two numbers whose product is -5 and whose sum is 4 . These numbers are -1 and 5 .

$$x^2 + 4x - 5 = 0$$

$$(x - 1)(x + 5) = 0$$

Therefore,

$$x - 1 = 0$$

or

$$x + 5 = 0$$

$$x = 1$$

$$x = -5$$

Therefore, the solutions are $x = 1$ and $x = -5$.

2. (a) Solve the quadratic equation $2x^2 + 7x + 3 = 0$ using the quadratic formula. Here, $a = 2$, $b = 7$ and $c = 3$. Substituting into the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-7 \pm \sqrt{7^2 - 4(2)(3)}}{2(2)}$$

$$= \frac{-7 \pm \sqrt{25}}{4}$$

$$= \frac{-7 \pm 5}{4}$$

Therefore,

$$x = \frac{-7 - 5}{4}$$

$$= \frac{-12}{4}$$

$$= -3$$

or

$$x = \frac{-7 + 5}{4}$$

$$= \frac{-2}{4}$$

$$= -\frac{1}{2}$$

Therefore, the solutions are $x = -3$ and $x = -\frac{1}{2}$.

- (b) Solve the quadratic equation $3x^2 - 2x - 8 = 0$ using the quadratic formula. Here, $a = 3$, $b = -2$ and $c = -8$. Substituting into the formula:

$$\begin{aligned}x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\x &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(3)(-8)}}{2(3)} \\&= \frac{2 \pm \sqrt{100}}{6} \\&= \frac{2 \pm 10}{6}\end{aligned}$$

Therefore,

$$\begin{aligned}x &= \frac{2 - 10}{6} \\&= \frac{-8}{6} \\&= -\frac{4}{3}\end{aligned}$$

or

$$\begin{aligned}x &= \frac{2 + 10}{6} \\&= \frac{12}{6} \\&= 2\end{aligned}$$

Therefore, the solutions are $x = -\frac{4}{3}$ and $x = 2$.

3. (a) Determine the nature of the roots of $x^2 + 2x + 5 = 0$. Using the discriminant

$$\Delta = b^2 - 4ac$$

with $a = 1$, $b = 2$ and $c = 5$, gives

$$\begin{aligned}\Delta &= 2^2 - 4(1)(5) \\&= 4 - 20 \\&= -16\end{aligned}$$

Since $\Delta < 0$, the quadratic has no real roots, i.e. the curve does not cross the x -axis.

- (b) Determine the nature of the roots of $x^2 - 6x + 9 = 0$. Using the discriminant

$$\Delta = b^2 - 4ac$$

with $a = 1$, $b = -6$ and $c = 9$, gives

$$\begin{aligned}\Delta &= (-6)^2 - 4(1)(9) \\ &= 36 - 36 \\ &= 0\end{aligned}$$

Since $\Delta = 0$, the quadratic has a repeated root, i.e. the curve crosses the x -axis once.

Solving:

$$\begin{aligned}x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(9)}}{2(1)} \\ &= \frac{6 \pm \sqrt{0}}{2} \\ &= \frac{6}{2} \\ &= 3\end{aligned}$$

Hence, $x = 3$. Therefore, the equation has a repeated root at $x = 3$.

- (c) Determine the nature of the roots of $2x^2 + x - 4 = 0$. Using the discriminant

$$\Delta = b^2 - 4ac$$

with $a = 2$, $b = 1$ and $c = -4$, gives

$$\begin{aligned}\Delta &= 1^2 - 4(2)(-4) \\ &= 1 + 32 \\ &= 33\end{aligned}$$

Since $\Delta > 0$, the quadratic has two real roots, i.e. the curve crosses the x -axis twice.

Solving:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-1 \pm \sqrt{1^2 - 4(2)(-4)}}{2(2)} \\ &= \frac{-1 \pm \sqrt{33}}{4} \end{aligned}$$

Hence, $x = -1.69$ and $x = 1.19$, both rounded to 2 decimal places.

4. (a) The ball reaches the ground at a height of zero metres, i.e $h = 0$. Therefore we wish to find t when $0 = -5t^2 + 20t + 25$. Using the quadratic formula gives:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-20 \pm \sqrt{20^2 - 4(-5)(25)}}{2(-5)} \\ &= \frac{-20 \pm \sqrt{900}}{-10} \\ &= \frac{-20 \pm 30}{-10} \end{aligned}$$

Therefore the solutions are $t = 5$ and $t = -1$. In terms of the context of the question, a negative time can be discounted. Thus, the ball reaches the ground at $t = 5$ seconds.

- (b) The sketch is below. Note that the y -intercept is at $(0, 25)$, which would be the initial height of the ball.

